

SPEC CPU 2026 Command Lines

Upasna^a, Venkata Kalyan Tavva^b

CASIG Lab, Indian Institute of Technology Ropar

^a*upasna.21csz0002@iitrpr.ac.in*

^b*kalyantv@iitrpr.ac.in*

SPEC CPU 2026 [1] Benchmark Suite contains total 52 benchmarks organized into four suites. SPECrate 2026 Integer (intrate), SPECSpeed 2026 Integer (intspeed), SPECrate 2026 Floating Point (fprate), and SPECSpeed 2026 Floating Point (fpspeed). Table 1 provides the workload-wise categorization across these four suites. There are three standard input sizes in SPEC 2026 benchmark suite: **test**, **train** and **ref** (**R**eference). To build and run the benchmarks, SPEC provides the **runcpu** utility. The command lines in this document are generated using **specinvoke -n** command executed in the respective **[benchmark]/run/[size]** directory.

Table 1: SPEC CPU 2026 Benchmarks

Sr No.	intrate	intspeed	fprate	fpspeed
1	706.stockfish_r	801.xz_s	709.cactus_r	800.pot3d_s
2	707.ntest_r	807.ntest_s	722.palm_r	803.sph_exa_s
3	708.sqlite_r	817.flac_s	731.ascenc_r	809.cactus_s
4	710.omnetpp_r	821.gcc_s	736.ocio_r	811.tealeaf_s
5	714.cpython_r	823.llvm_s	737.gmsh_r	816.nab_s
6	721.gcc_r	827.cppcheck_s	748.flightdm_r	820.cloverleaf_s
7	723.llvm_r	829.abc_s	749.fotonik3d_r	822.palm_s
8	727.cppcheck_r	834.vpr_s	765.roms_r	849.fotonik3d_s
9	729.abc_r	835.gem5_s	766.femflow_r	857.namd_s
10	734.vpr_r	838.diamond_s	767.nest_r	865.roms_s
11	735.gem5_r	846.minizinc_s	772.marian_r	867.nest_s
12	750.sealcrypto_r	853.ns3_s	782.lbm_r	872.marian_s
13	753.ns3_r	854.graph500_s		881.neutron_s
14	777.zstd_r			

SPEC CPU Rate 2026 Integer Benchmarks

1. 706.stockfish_r

1.1. Test

1. stockfish_base.mytest bench 16 1 15 spec_test.fen depth nnue >
stockfish_spec_test_nnue.out 2>> stockfish_spec_test_nnue.err
2. stockfish_base.mytest bench 16 1 15 spec_test.fen depth classical >
stockfish_spec_test_classical.out 2>> stockfish_spec_test_classical.err

1.2. Train

1. stockfish_base.mytest bench 1600 1 24 spec_train.fen depth nnue >
stockfish_spec_train_nnue.out 2>> stockfish_spec_train_nnue.err
2. stockfish_base.mytest bench 1600 1 24 spec_train.fen depth classical >
stockfish_spec_train_classical.out 2>> stockfish_spec_train_classical.err

1.3. Ref

1. stockfish_base.mytest bench 1600 1 26 spec_ref_pos_1to6.fen depth classical >
stockfish_spec_ref_pos_1to6_classical.out 2>>
stockfish_spec_ref_pos_1to6_classical.err
2. stockfish_base.mytest bench 1600 1 26 spec_ref_pos_1to6.fen depth nnue >
stockfish_spec_ref_pos_1to6_nnue.out 2>> stockfish_spec_ref_pos_1to6_nnue.err
3. stockfish_base.mytest bench 1600 1 26 spec_ref_pos_7to11.fen depth nnue >
stockfish_spec_ref_pos_7to11_nnue.out 2>> stockfish_spec_ref_pos_7to11_nnue.err

2. 707.ntest_r

2.1. Test

1. ntest_r_base.mytest OneGame.ggf 10 6 > ntest.OneGame.10.6.out 2>>
ntest.OneGame.10.6.err

2.2. Train

1. ntest_r_base.mytest Othello.60e4.ggf 19 19 > ntest.Othello.60e4.19.19.out 2>>
ntest.Othello.60e4.19.19.err

2.3. Ref

```
1. ntest_r_base.mytest 0thello.154.ggf 20 16 > ntest.0thello.154.20.16.out 2>>
   ntest.0thello.154.20.16.err
```

3. 708.sqlite_r

3.1. Test

```
1. sqlite_r_base.mytest --memdb --size 10 --testset main --verify >
   sqlite_r.main.out 2>> sqlite_r.main.err
```

3.2. Train

```
1. sqlite_r_base.mytest --memdb --size 500 --testset main --verify >
   sqlite_r.main.out 2>> sqlite_r.main.err

2. sqlite_r_base.mytest --memdb --size 500 --testset cte --verify >
   sqlite_r.cte.out 2>> sqlite_r.cte.err
```

3.3. Ref

```
1. sqlite_r_base.mytest --memdb --size 2000 --testset main --verify >
   sqlite_r.main.out 2>> sqlite_r.main.err

2. sqlite_r_base.mytest --memdb --size 2000 --testset cte --verify >
   sqlite_r.cte.out 2>> sqlite_r.cte.err

3. sqlite_r_base.mytest --memdb --size 1000 --testset fp --verify >
   sqlite_r.fp.out 2>> sqlite_r.fp.err
```

4. 710.omnetpp_r

4.1. Test

```
1. omnetpp_r_base.mytest -f randomMesh.ini -c General > omnetpp.General-0.out 2>>
   omnetpp.General-0.err

2. omnetpp_r_base.mytest -f queuenet.ini -c OneFifo > omnetpp.OneFifo-0.out 2>>
   omnetpp.OneFifo-0.err

3. omnetpp_r_base.mytest -f queuenet.ini -c AllocDealloc >
   omnetpp.AllocDealloc-0.out 2>> omnetpp.AllocDealloc-0.err
```

4.2. Train

1. `omnetpp_r_base.mytest -f randomMesh.ini -c General > omnetpp.General-0.out 2>> omnetpp.General-0.err`
2. `omnetpp_r_base.mytest -f queuenet.ini -c OneFifo > omnetpp.OneFifo-0.out 2>> omnetpp.OneFifo-0.err`

4.3. Ref

1. `omnetpp_r_base.mytest -f randomMesh.ini -c General > omnetpp.General-0.out 2>> omnetpp.General-0.err`
2. `omnetpp_r_base.mytest -f queuenet.ini -c OneFifo > omnetpp.OneFifo-0.out 2>> omnetpp.OneFifo-0.err`
3. `omnetpp_r_base.mytest -f queuenet.ini -c TandemFifos > omnetpp.TandemFifos-0.out 2>> omnetpp.TandemFifos-0.err`

5. 714.cpython_r

5.1. Test

1. `cpython_r_base.mytest -I -B coreml_pb.py -i 2 -d 20 > cpython_r_0.out 2>> cpython_r_0.err`

5.2. Train

1. `cpython_r_base.mytest -I -B coreml_pb.py -i 10 -a -c -m PoseNetMobileNet100S16FP16.mlmodel > cpython_r_0.out 2>> cpython_r_0.err`
2. `cpython_r_base.mytest -I -B dna_bench.py 200000 > cpython_r_1.out 2>> cpython_r_1.err`

5.3. Ref

1. `cpython_r_base.mytest -I -B coreml_pb.py -i 2 -a -m Resnet50Headless.mlmodel -d 10 > cpython_r_0.out 2>> cpython_r_0.err`
2. `cpython_r_base.mytest -I -B coreml_pb.py -i 5 -a -c -m MobileNetV2.mlmodel -d 20 > cpython_r_1.out 2>> cpython_r_1.err`
3. `cpython_r_base.mytest -I -B dna_bench.py 600000 > cpython_r_2.out 2>> cpython_r_2.err`

6. 721.gcc_r

6.1. Test

```
1. cc1_r_base.mytest t1.c -O3 -finline-limit=50000 -o
   t1.c.opts-O3_-finline-limit_50000.s > t1.c.opts-O3_-finline-limit_50000.out 2>>
   t1.c.opts-O3_-finline-limit_50000.err
```

6.2. Train

```
1. cc1_r_base.mytest 200.c -O2 -o 200.c.opts-O2.s > 200.c.opts-O2.out 2>>
   200.c.opts-O2.err

2. cc1_r_base.mytest scilab.c -O3 -o scilab.c.opts-O3.s > scilab.c.opts-O3.out 2>>
   scilab.c.opts-O3.err

3. cc1_r_base.mytest train01.c -O3 -finline-limit=50000 -o
   train01.c.opts-O3_-finline-limit_50000.s >
   train01.c.opts-O3_-finline-limit_50000.out 2>>
   train01.c.opts-O3_-finline-limit_50000.err
```

6.3. Ref

```
1. cc1_r_base.mytest gcc-pp.c -O2 -fpic -o gcc-pp.c.opts-O2_-fpic.s >
   gcc-pp.c.opts-O2_-fpic.out 2>> gcc-pp.c.opts-O2_-fpic.err

2. cc1_r_base.mytest gcc-smaller.c -O3 -fipa-pta -o
   gcc-smaller.c.opts-O3_-fipa-pta.s > gcc-smaller.c.opts-O3_-fipa-pta.out 2>>
   gcc-smaller.c.opts-O3_-fipa-pta.err

3. cc1_r_base.mytest ref32.c -O3 -finline-limit=12000 -fno-tree-vrp -o
   ref32.c.opts-O3_-finline-limit_12000_-fno-tree-vrp.s >
   ref32.c.opts-O3_-finline-limit_12000_-fno-tree-vrp.out 2>>
   ref32.c.opts-O3_-finline-limit_12000_-fno-tree-vrp.err
```

7. 723.llvm_r

7.1. Test

```
1. llvm-opt_r_base.mytest test.ll -S -O3 -mcpu=pwr9 --sha512 -o
   test.ll.opts-S_-O3_-mcpu_pwr9.ll > test.ll.opts-S_-O3_-mcpu_pwr9.out 2>>
   test.ll.opts-S_-O3_-mcpu_pwr9.err
```

7.2. Train

1. `llvm-opt_r_base.mytest train.ll -S -O3 -mcpu=pwr9 --sha512 -o train.ll.opts-S_-O3_-mcpu_pwr9.ll > train.ll.opts-S_-O3_-mcpu_pwr9.out 2>> train.ll.opts-S_-O3_-mcpu_pwr9.err`
2. `llvm-opt_r_base.mytest ELFObject.ll -S -O2 -inline-threshold=50000 -mcpu=pwr9 --sha512 -o ELFObject.ll.opts-S_-O2_-inline-threshold_50000_-mcpu_pwr9.ll > ELFObject.ll.opts-S_-O2_-inline-threshold_50000_-mcpu_pwr9.out 2>> ELFObject.ll.opts-S_-O2_-inline-threshold_50000_-mcpu_pwr9.err`

7.3. Ref

1. `llvm-opt_r_base.mytest transformspplus.bc -S -O3 -mcpu=pwr9 --sha512 -o transformspplus.bc.opts-S_-O3_-mcpu_pwr9.ll > transformspplus.bc.opts-S_-O3_-mcpu_pwr9.out 2>> transformspplus.bc.opts-S_-O3_-mcpu_pwr9.err`
2. `llvm-opt_r_base.mytest codegen.bc -S -O3 -mcpu=pwr9 --sha512 -o codegen.bc.opts-S_-O3_-mcpu_pwr9.ll > codegen.bc.opts-S_-O3_-mcpu_pwr9.out 2>> codegen.bc.opts-S_-O3_-mcpu_pwr9.err`

8. 727.cppcheck_r

8.1. Test

1. `testrunner_r_base.mytest > testrunner_r.out 2>> testrunner_r.err`
2. `cppcheck_r_base.mytest --force hello.c --output-file=hello_bogey.txt --platform=unix64 > cppcheck_r.hello_bogey.out 2>> cppcheck_r.hello_bogey.err`

8.2. Train

1. `cppcheck_r_base.mytest --force lua-5.4.7/lcode.c --enable=all --checkers-report=lcode_report.txt --output-file=lcode_bogey.txt --suppress=missingIncludeSystem --platform=unix64 > cppcheck_r.lcode_bogey.out 2>> cppcheck_r.lcode_bogey.err`
2. `cppcheck_r_base.mytest --force automaticMeshSizeField.cpp meshGRegionBoundaryLayer.cpp meshGRegionLocalMeshMod.cpp --enable=all --checkers-report=mesh_report.txt --output-file=mesh_bogey.txt --enable=all --suppress=missingIncludeSystem --suppress=missingInclude --check-level=exhaustive --platform=unix64 > cppcheck_r.mesh_bogey.out 2>> cppcheck_r.mesh_bogey.err`

8.3. Ref

```

1. cppcheck_r_base.mytest --force 738-diamond-record.cpp
   --checkers-report=738_report.txt --enable=all --output-file=738_bogey.txt
   --platform=unix64 > cppcheck_r.738_bogey.out 2>> cppcheck_r.738_bogey.err

2. cppcheck_r_base.mytest --force 747-dealii-data_out_base.cc
   --checkers-report=747_report.txt --enable=all --output-file=747_bogey.txt
   --platform=unix64 > cppcheck_r.747_bogey.out 2>> cppcheck_r.747_bogey.err

3. cppcheck_r_base.mytest --force 770-7z-SystemPage.cpp
   --checkers-report=770_report.txt --output-file=770_bogey.txt --platform=unix64
   > cppcheck_r.770_bogey.out 2>> cppcheck_r.770_bogey.err

```

9. 729.abc_r

9.1. Test

```

1. abc_r_base.mytest -F truthtable.in > truthtable.out 2>> truthtable.err
2. abc_r_base.mytest -F riscv.in > riscv.out 2>> riscv.err

```

9.2. Train

```

1. abc_r_base.mytest -F ucsb_152.in > ucsb_152.out 2>> ucsb_152.err
2. abc_r_base.mytest -F des_perf.in > des_perf.out 2>> des_perf.err
3. abc_r_base.mytest -F sparcT1_core.in > sparcT1_core.out 2>> sparcT1_core.err

```

9.3. Ref

```

1. abc_r_base.mytest -F twoexact.in > twoexact.out 2>> twoexact.err
2. abc_r_base.mytest -F beem6-fraig.in > beem6-fraig.out 2>> beem6-fraig.err
3. abc_r_base.mytest -F mem_ctrl.in > mem_ctrl.out 2>> mem_ctrl.err
4. abc_r_base.mytest -F vga_lcd_miter.in > vga_lcd_miter.out 2>> vga_lcd_miter.err
5. abc_r_base.mytest -F mcml.in > mcml.out 2>> mcml.err

```

10. 734.vpr_r

10.1. Test

1. `vpr_base.mytest stratixiv_arch.timing.xml murax_stratixiv_arch_timing.blif`
`--place_algorithm bounding_box --RL_agent_placement off --max_criticality 0.0`
`--init_t 512 --alpha_t 0.75 --exit_t 1 --router_initial_timing all_critical`
`--routing_failure_predictor off --route_chan_width 300 --max_router_iterations`
`20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0`
`--astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file`
`murax_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0`
`--base_cost_type demand_only --inner_num 4 --read_initial_place_file`
`ref_murax_stratixiv_arch_timing.init.place --place >`
`murax_stratixiv_arch_timing.blif.place.out 2>>`
`murax_stratixiv_arch_timing.blif.place.err`
2. `vpr_base.mytest stratixiv_arch.timing.xml murax_stratixiv_arch_timing.blif`
`--place_algorithm bounding_box --place_static_notiming_move_prob 50 25 25`
`--max_criticality 0.0 --router_initial_timing all_critical`
`--routing_failure_predictor off --route_chan_width 300 --max_router_iterations`
`20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0`
`--astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file`
`murax_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0`
`--place_file ref_murax_stratixiv_arch_timing.place --base_cost_type demand_only`
`--analysis --route > murax_stratixiv_arch_timing.blif.route.out 2>>`
`murax_stratixiv_arch_timing.blif.route.err`

10.2. Train

```

1. vpr_base.mytest stratixiv_arch.timing.xml
   sudoku_check_stratixiv_arch_timing.blif --place_algorithm bounding_box
   --RL_agent_placement off --max_criticality 0.0 --init_t 512 --alpha_t 0.75
   --exit_t 1 --router_initial_timing all_critical --routing_failure_predictor off
   --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
   --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
   --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sudoku_check_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --inner_num 4 --read_initial_place_file
   ref_sudoku_check_stratixiv_arch_timing.init.place --place >
   sudoku_check_stratixiv_arch_timing.blif.place.out 2>>
   sudoku_check_stratixiv_arch_timing.blif.place.err

2. vpr_base.mytest stratixiv_arch.timing.xml
   sudoku_check_stratixiv_arch_timing.blif --place_algorithm bounding_box
   --place_static_notiming_move_prob 50 25 25 --max_criticality 0.0
   --router_initial_timing all_critical --routing_failure_predictor off
   --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
   --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
   --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sudoku_check_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --place_file
   ref_sudoku_check_stratixiv_arch_timing.place --analysis --route >
   sudoku_check_stratixiv_arch_timing.blif.route.out 2>>
   sudoku_check_stratixiv_arch_timing.blif.route.err

```

10.3. Ref

1.

```
vpr_base.mytest stratixiv_arch.timing.xml JPEG_stratixiv_arch_timing.blif
--RL_agent_placement off --place_algorithm bounding_box --max_criticality 0.0
--init_t 512 --alpha_t 0.75 --exit_t 1 --router_initial_timing all_critical
--routing_failure_predictor off --route_chan_width 300 --max_router_iterations
20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0
--astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
JPEG_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
--base_cost_type demand_only --inner_num 4 --read_initial_place_file
ref_JPEG_stratixiv_arch_timing.init.place --place >
JPEG_stratixiv_arch_timing.blif.place.out 2>>
JPEG_stratixiv_arch_timing.blif.place.err
```
2.

```
vpr_base.mytest stratixiv_arch.timing.xml JPEG_stratixiv_arch_timing.blif
--place_algorithm bounding_box --place_static_notiming_move_prob 50 25 25
--max_criticality 0.0 --router_initial_timing all_critical
--routing_failure_predictor off --route_chan_width 300 --max_router_iterations
20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0
--astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
JPEG_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
--base_cost_type demand_only --place_file ref_JPEG_stratixiv_arch_timing.place
--analysis --route > JPEG_stratixiv_arch_timing.blif.route.out 2>>
JPEG_stratixiv_arch_timing.blif.route.err
```
3.

```
vpr_base.mytest stratixiv_arch.timing.xml
smithwaterman_stratixiv_arch_timing.blif --RL_agent_placement off
--place_algorithm bounding_box --max_criticality 0.0 --init_t 512 --alpha_t
0.75 --exit_t 1 --router_initial_timing all_critical
--routing_failure_predictor off --route_chan_width 300 --max_router_iterations
20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0
--astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
smithwaterman_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity
0 --base_cost_type demand_only --inner_num 1.8 --read_initial_place_file
ref_smithwaterman_stratixiv_arch_timing.init.place --place >
smithwaterman_stratixiv_arch_timing.blif.place.out 2>>
smithwaterman_stratixiv_arch_timing.blif.place.err
```
4.

```
vpr_base.mytest stratixiv_arch.timing.xml
smithwaterman_stratixiv_arch_timing.blif --place_algorithm bounding_box
--place_static_notiming_move_prob 50 25 25 --max_criticality 0.0
--router_initial_timing all_critical --routing_failure_predictor off
--route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
--initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
--router_profiler_astar_fac 1.5 --seed 3 --sdc_file
smithwaterman_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity
0 --base_cost_type demand_only --place_file
ref_smithwaterman_stratixiv_arch_timing.place --analysis --route >
smithwaterman_stratixiv_arch_timing.blif.route.out 2>>
smithwaterman_stratixiv_arch_timing.blif.route.err
```

11. 735.gem5_r

11.1. Test

```
1. gem5sim_base.mytest
  --stats-file=synthetic_traffic.py_LinearGenerator_100.stats.txt
  synthetic_traffic.py LinearGenerator 100 >
  synthetic_traffic.py_LinearGenerator_100.out 2>>
  synthetic_traffic.py_LinearGenerator_100.err
```

11.2. Train

```
1. gem5sim_base.mytest
  --stats-file=synthetic_traffic.py_LinearGenerator_100.stats.txt
  synthetic_traffic.py LinearGenerator 100 >
  synthetic_traffic.py_LinearGenerator_100.out 2>>
  synthetic_traffic.py_LinearGenerator_100.err
```

11.3. Ref

```
1. gem5sim_base.mytest
  --stats-file=run_riscv_boot.py_o3_10_--max-ticks_10_000_000_000.stats.txt
  run_riscv_boot.py o3 10 --max-ticks 10_000_000_000 >
  run_riscv_boot.py_o3_10_--max-ticks_10_000_000_000.out 2>>
  run_riscv_boot.py_o3_10_--max-ticks_10_000_000_000.err

2. gem5sim_base.mytest
  --stats-file=run_riscv_boot.py_timing_4_--max-ticks_20_000_000_000.stats.txt
  run_riscv_boot.py timing 4 --max-ticks 20_000_000_000 >
  run_riscv_boot.py_timing_4_--max-ticks_20_000_000_000.out 2>>
  run_riscv_boot.py_timing_4_--max-ticks_20_000_000_000.err

3. gem5sim_base.mytest
  --stats-file=synthetic_traffic.py_LinearGenerator_21.stats.txt
  synthetic_traffic.py LinearGenerator 21 >
  synthetic_traffic.py_LinearGenerator_21.out 2>>
  synthetic_traffic.py_LinearGenerator_21.err

4. gem5sim_base.mytest
  --stats-file=synthetic_traffic.py_LinearGenerator_74_--ruby.stats.txt
  synthetic_traffic.py LinearGenerator 74 --ruby >
  synthetic_traffic.py_LinearGenerator_74_--ruby.out 2>>
  synthetic_traffic.py_LinearGenerator_74_--ruby.err
```

12. 750.sealcrypto_r

12.1. Test

```
1. sealcrypto_r_base.mytest test country_capital_test.csv Ecuador >
   homoencrypt.test_country_capital_test.csv_Ecuador.out 2>>
   homoencrypt.test_country_capital_test.csv_Ecuador.err
```

12.2. Train

```
1. sealcrypto_r_base.mytest train australia_state_capitals_train.csv Victoria >
   homoencrypt.train_australia_state_capitals_train.csv_Victoria.out 2>>
   homoencrypt.train_australia_state_capitals_train.csv_Victoria.err
```

12.3. Ref

```
1. sealcrypto_r_base.mytest refrate ecuador_province_capitals_refrate.csv
   Galapagos >
   homoencrypt.refrate_ecuador_province_capitals_refrate.csv_Galapagos.out 2>>
   homoencrypt.refrate_ecuador_province_capitals_refrate.csv_Galapagos.err
```

13. 753.ns3_r

13.1. Test

```
1. ns3_r_base.mytest tcp-pacing --isPacingEnabled=true --tracing=false --RngSeed=1
   --RngRun=1 > tcp-pacing_0.out 2>> tcp-pacing_0.err

2. ns3_r_base.mytest wifi-mixed-network --simulationTime=2 --RngSeed=1 --RngRun=1
   > wifi-mixed-network_1.out 2>> wifi-mixed-network_1.err
```

13.2. Train

```
1. ns3_r_base.mytest wifi-mixed-network --isUdp=0 --simulationTime=10 --RngSeed=1
   --RngRun=1 > wifi-mixed-network_0.out 2>> wifi-mixed-network_0.err

2. ns3_r_base.mytest wifi-mixed-network --isUdp=1 --simulationTime=10
   --payloadSize=2048 --RngSeed=1 --RngRun=1 > wifi-mixed-network_1.out 2>>
   wifi-mixed-network_1.err
```

13.3. Ref

```
1. ns3_r_base.mytest mobile-scenario --simTimeMinutes=3 --RngSeed=1 --RngRun=1 >
   mobile-scenario_0.out 2>> mobile-scenario_0.err

2. ns3_r_base.mytest tcp-pacing --simulationEndTime=500 --useEcn=false --RngSeed=1
   --RngRun=1 > tcp-pacing_1.out 2>> tcp-pacing_1.err

3. ns3_r_base.mytest lena-radio-link-failure --numberOfEnbs=2
   --interSiteDistance=800 --simTime=200 --RngSeed=1 --RngRun=1 >
   lena-radio-link-failure_2.out 2>> lena-radio-link-failure_2.err
```

14. 777.zstd.r

14.1. Test

```
1. zstd_base.mytest -b3 --verbose -i1 > zstd.b3_verbose_i1.out 2>>
   zstd.b3_verbose_i1.err
```

14.2. Train

```
1. zstd_base.mytest -b3 -e12 --verbose -i15 > zstd.b3_e12_verbose_i15.out 2>>
   zstd.b3_e12_verbose_i15.err
```

14.3. Ref

```
1. zstd_base.mytest -b3 -e3 --verbose -i40 cld.tar >
   zstd.b3_e3_verbose_i40_cld.tar.out 2>> zstd.b3_e3_verbose_i40_cld.tar.err

2. zstd_base.mytest -b5 -e5 --verbose -i25 cld.tar >
   zstd.b5_e5_verbose_i25_cld.tar.out 2>> zstd.b5_e5_verbose_i25_cld.tar.err

3. zstd_base.mytest -b7 -e7 --verbose -i12 cld.tar >
   zstd.b7_e7_verbose_i12_cld.tar.out 2>> zstd.b7_e7_verbose_i12_cld.tar.err
```

SPEC CPU Speed 2026 Integer Benchmarks

1. 801.xz_s

1.1. Test

```

1. xz_s_base.mytest cpu2006docs.tar.xz 4
   055ce243071129412e9dd0b3b69a21654033a9b723d874b2015c774fac1553d9713be561ca86f74e4f
   16f22e664fc17a79f30caa5ad2c04fbc447549c2810fae 1548636 1555348 0 >
   cpu2006docs.tar-4-0.out 2>> cpu2006docs.tar-4-0.err

2. xz_s_base.mytest cpu2006docs.tar.xz 4
   055ce243071129412e9dd0b3b69a21654033a9b723d874b2015c774fac1553d9713be561ca86f74e4f
   16f22e664fc17a79f30caa5ad2c04fbc447549c2810fae 1462248 1467400 1 >
   cpu2006docs.tar-4-1.out 2>> cpu2006docs.tar-4-1.err

3. xz_s_base.mytest cpu2006docs.tar.xz 4
   055ce243071129412e9dd0b3b69a21654033a9b723d874b2015c774fac1553d9713be561ca86f74e4f
   16f22e664fc17a79f30caa5ad2c04fbc447549c2810fae 1428548 1428556 2 >
   cpu2006docs.tar-4-2.out 2>> cpu2006docs.tar-4-2.err

```

1.2. Train

```

1. xz_s_base.mytest input.combined.xz 40
   a841f68f38572a49d86226b7ff5baeb31bd19dc637a922a972b2e6d1257a890f6a544ecab967c313e3
   70478c74f760eb229d4eef8a8d2836d233d3e9dd1430bf 6356684 6356692 8 >
   input.combined-40-8.out 2>> input.combined-40-8.err

2. xz_s_base.mytest IMG_2560.cr2.xz 40
   ec03e53b02deae89b6650f1de4bed76a012366fb3d4bdc791e8633d1a5964e03004523752ab008eff0
   d9e693689c53056533a05fc4b277f0086544c6c3cbbbb6 40822692 40827404 4 >
   IMG_2560.cr2-40-4.out 2>> IMG_2560.cr2-40-4.err

```

1.3. Ref

```

1. xz_s_base.mytest cpu2006docs.tar.xz 15000
   055ce243071129412e9dd0b3b69a21654033a9b723d874b2015c774fac1553d9713be561ca86f74e4f
   16f22e664fc17a79f30caa5ad2c04fbc447549c2810fae 2321296421 2523148284 4 >
   cpu2006docs.tar-15000-4.out 2>> cpu2006docs.tar-15000-4.err

2. xz_s_base.mytest cld.tar.xz 2700
   19cf30ae51eddcbefda78dd06014b4b96281456e078ca7c13e1c0c9e6a8d8ff3efb4ad6b04566977
   18cede6bd5454852652806a657bb56e07d61128434b474 959312906 1042731420 8 >
   cld.tar-2700-8.out 2>> cld.tar-2700-8.err

```

2. 807.ntest_s

2.1. Test

```
1. ntest_s_base.mytest OneGame.ggf 10 6 > ntest.OneGame.10.6.out 2>>
   ntest.OneGame.10.6.err
```

2.2. Train

```
1. ntest_s_base.mytest 0thello.60e4.ggf 19 19 > ntest.0thello.60e4.19.19.out 2>>
   ntest.0thello.60e4.19.19.err
```

2.3. Ref

```
1. ntest_s_base.mytest 0thello.01e4.ggf 20 21 > ntest.0thello.01e4.20.21.out 2>>
   ntest.0thello.01e4.20.21.err
```

3. 817.flac_s

3.1. Test

```
1. flac_s_base.mytest -7V -f amp_podcast_11.wav -o amp_podcast_11.flac
   --no-preserve-modtime -j 4 > flac.amp_podcast_11.flac.out 2>>
   flac.amp_podcast_11.flac.err

2. flac_s_base.mytest -a -f amp_podcast_11.flac -o amp_podcast_11.analysis.txt
   --no-preserve-modtime --delete-input-file -j 4 >
   flac.amp_podcast_11.analysis.txt.out 2>> flac.amp_podcast_11.analysis.txt.err
```

3.2. Train

```
1. flac_s_base.mytest -8Ve -f git_etiquette_mono.wav -o git_etiquette_mono.flac
   --keep-foreign-metadata --no-preserve-modtime -j 4 >
   flac.git_etiquette_mono.flac.out 2>> flac.git_etiquette_mono.flac.err

2. flac_s_base.mytest -t -f git_etiquette_mono.flac -o git_etiquette_test
   --delete-input-file -j 4 > flac.git_etiquette_test.out 2>>
   flac.git_etiquette_test.err
```

3.3. Ref

```

1. flac_s_base.mytest -8Vep -f learning_is_the_new_literacy.wav -o
   learning_is_the_new_literacy1.flac --mid-side --replay-gain
   --keep-foreign-metadata --no-preserve-modtime -j 4 >
   flac.learning_is_the_new_literacy1.flac.out 2>>
   flac.learning_is_the_new_literacy1.flac.err

2. flac_s_base.mytest -7Vep -f learning_is_the_new_literacy.wav -o
   learning_is_the_new_literacy2.flac --mid-side --replay-gain
   --keep-foreign-metadata --no-preserve-modtime -j 4 >
   flac.learning_is_the_new_literacy2.flac.out 2>>
   flac.learning_is_the_new_literacy2.flac.err

3. flac_s_base.mytest -6Vep -f learning_is_the_new_literacy.wav -o
   learning_is_the_new_literacy3.flac --replay-gain --adaptive-mid-side
   --keep-foreign-metadata --no-preserve-modtime -j 4 >
   flac.learning_is_the_new_literacy3.flac.out 2>>
   flac.learning_is_the_new_literacy3.flac.err

```

4. 821.gcc_s

4.1. Test

```

1. cc1_s_base.mytest t1.c -O3 -finline-limit=50000 -o t1.c.0001.s
2. cc1_s_base.mytest t1.c -O2 -finline-limit=50000 -w -o t1.c.0002.s
3. cc1_s_base.mytest t1.c -O1 -finline-limit=50000 -w -o t1.c.0003.s
4. cc1_s_base.mytest t1.c -Os -finline-limit=50000 -o t1.c.0004.s

```

4.2. Train

```

1. cc1_s_base.mytest 200.c -O3 -finline-limit=50000 -o 200.c.0001.s
2. cc1_s_base.mytest scilab.c -O3 -finline-limit=50000 -o scilab.c.0001.s
3. cc1_s_base.mytest train01.c -O3 -finline-limit=50000
   -Wno-builtin-declaration-mismatch -o train01.c.0002.s
4. cc1_s_base.mytest 200.c -O2 -finline-limit=50000 -o 200.c.0003.s
5. cc1_s_base.mytest scilab.c -O2 -finline-limit=50000 -o scilab.c.0003.s

```


4.3. Ref

1. `cc1_s_base.mytest 721.gcc/mpfr/sqr.c -w -Ofast -finline-limit=24000 -fgcse -fgcse-las -fgcse-lm -fgcse-sm -Wno-switch-bool -Wno-stringop-overflow -o 721.gcc_mpfr_sqr.c.0013.s`
2. `cc1_s_base.mytest 735.gem5/cpython/Objects/unicodeobject.c -O3 -finline-limit=20000 -falign-functions -falign-jumps -falign-labels -falign-loops -fgcse-sm -w -o 735.gem5_cpython_Objects_unicodeobject.c.1251.s`
3. `cc1_s_base.mytest 709.cactus/repos/cactusbase/Boundary/src/ScalarBoundary.c -O2 -finline-limit=20000 -funroll-loops -fselective-scheduling -w -o 709.cactus_repos_cactusbase_Boundary_src_ScalarBoundary.c.0444.s`

5. 823.llvm.s

5.1. Test

1. `llvm_s_base.mytest test.ll -S -O1 --sha512 -o test.ll.0001.sha512`
2. `llvm_s_base.mytest test.ll -S -O2 --sha512 -o test.ll.0002.sha512`
3. `llvm_s_base.mytest test.ll -S -O3 --sha512 -o test.ll.0003.sha512`
4. `llvm_s_base.mytest test.ll -S -Os --sha512 -o test.ll.0004.sha512`
5. `llvm_s_base.mytest test.ll -S -Oz --sha512 -o test.ll.0005.sha512`

5.2. Train

1. `llvm_s_base.mytest train.ll -S -O1 --sha512 -o train.ll.0001.sha512`
2. `llvm_s_base.mytest train.ll -S -O2 --sha512 -o train.ll.0002.sha512`
3. `llvm_s_base.mytest train.ll -S -O3 --sha512 -o train.ll.0003.sha512`
4. `llvm_s_base.mytest train.ll -S -Os --sha512 -o train.ll.0004.sha512`
5. `llvm_s_base.mytest train.ll -S -Oz --sha512 -o train.ll.0005.sha512`

5.3. Ref

1. `llvm_s_base.mytest 747/particle_handler.ll -S -O3 -inline-threshold=1000 --slp-vectorizer --reg2mem --sha512 -o 747_particle_handler.ll.0180.sha512`
2. `llvm_s_base.mytest 747/vector_tools_project.ll -S -Oz -inline-threshold=40000 --slp-vectorizer --sink --reg2mem --sha512 -o 747_vector_tools_project.ll.0141.sha512`
3. `llvm_s_base.mytest 747/dof_tools_sparsity.ll -S -Oz -inline-threshold=1000 --scalarizer --sink --sha512 -o 747_dof_tools_sparsity.ll.0052.sha512`

6. 827.cppcheck_s

6.1. Test

```
1. testrunner_s_base.mytest > testrunner_s.out 2>> testrunner_s.err
2. cppcheck_s_base.mytest --file-list=check_file_list --cppcheck-build-dir=. -j 4
   --platform=unix64 > cppcheck_s.raw.out 2>> cppcheck_s.raw.err
```

6.2. Train

```
1. cppcheck_s_base.mytest --cppcheck-build-dir=lua-5.4.7 --enable=all
   --disable=unusedFunction --suppress=toomanyconfigs
   --suppress=missingIncludeSystem --checkers-report=lua_report.txt lua-5.4.7 -j 4
   --platform=unix64 > cppcheck_s.raw.out 2>> cppcheck_s.raw.err
```

6.3. Ref

```
1. cppcheck_s_base.mytest --file-list=check_file_list -f
   --suppress=normalCheckLevelMaxBranches --checkers-report=report.txt
   --cppcheck-build-dir=src --disable=unusedFunction -j 4 --platform=unix64 >
   cppcheck_s.raw.out 2>> cppcheck_s.raw.err
```

7. 829.abc_s

7.1. Test

```
1. abc_base.mytest -F truthtable.in > truthtable.out 2>> truthtable.err
2. abc_base.mytest -F riscv.in > riscv.out 2>> riscv.err
```

7.2. Train

```
1. abc_base.mytest -F ucsb_152.in > ucsb_152.out 2>> ucsb_152.err
2. abc_base.mytest -F des_perf.in > des_perf.out 2>> des_perf.err
3. abc_base.mytest -F sparct1_core.in > sparct1_core.out 2>> sparct1_core.err
```

7.3. Ref

1. `abc_base.mytest -F ethernet-fraig.in > ethernet-fraig.out 2>> ethernet-fraig.err`
2. `abc_base.mytest -F bitcoin.in > bitcoin.out 2>> bitcoin.err`

8. 834.vpr_s

8.1. Test

1. `vpr_base.mytest stratixiv_arch.timing.xml murax_stratixiv_arch_timing.blif --place_algorithm bounding_box --RL_agent_placement off --max_criticality 0.0 --init_t 512 --alpha_t 0.75 --exit_t 1 --router_initial_timing all_critical --routing_failure_predictor off --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file murax_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0 --base_cost_type demand_only --inner_num 4 --read_initial_place_file ref_murax_stratixiv_arch_timing.init.place --place > murax_stratixiv_arch_timing.blif.out 2>> murax_stratixiv_arch_timing.blif.err`
2. `vpr_base.mytest stratixiv_arch.timing.xml murax_stratixiv_arch_timing.blif --place_algorithm bounding_box --place_static_notiming_move_prob 50 25 25 --max_criticality 0.0 --router_initial_timing all_critical --routing_failure_predictor off --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file murax_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0 --place_file ref_murax_stratixiv_arch_timing.place --base_cost_type demand_only --analysis --route > murax_stratixiv_arch_timing.blif.out 2>> murax_stratixiv_arch_timing.blif.err`

8.2. *Train*

```

1. vpr_base.mytest stratixiv_arch.timing.xml
   sudoku_check_stratixiv_arch_timing.blif --place_algorithm bounding_box
   --RL_agent_placement off --max_criticality 0.0 --init_t 512 --alpha_t 0.75
   --exit_t 1 --router_initial_timing all_critical --routing_failure_predictor off
   --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
   --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
   --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sudoku_check_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --inner_num 4 --read_initial_place_file
   ref_sudoku_check_stratixiv_arch_timing.init.place --place >
   sudoku_check_stratixiv_arch_timing.blif.out 2>>
   sudoku_check_stratixiv_arch_timing.blif.err

2. vpr_base.mytest stratixiv_arch.timing.xml
   sudoku_check_stratixiv_arch_timing.blif --place_algorithm bounding_box
   --place_static_notiming_move_prob 50 25 25 --max_criticality 0.0
   --router_initial_timing all_critical --routing_failure_predictor off
   --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
   --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
   --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sudoku_check_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --place_file
   ref_sudoku_check_stratixiv_arch_timing.place --analysis --route >
   sudoku_check_stratixiv_arch_timing.blif.out 2>>
   sudoku_check_stratixiv_arch_timing.blif.err

```

8.3. Ref

```

1. vpr_base.mytest stratixiv_arch.timing.xml
   sparct1_core_stratixiv_arch_timing.blif --RL_agent_placement off
   --place_algorithm bounding_box --max_criticality 0.0 --init_t 512 --alpha_t
   0.75 --exit_t 1 --router_initial_timing all_critical
   --routing_failure_predictor off --route_chan_width 300 --max_router_iterations
   20 --router_lookahead classic --initial_pres_fac 1.0 --pres_fac_mult 2.0
   --astar_fac 1.5 --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sparct1_core_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --inner_num 3 --read_initial_place_file
   ref_sparct1_core_stratixiv_arch_timing.init.place --place >
   sparct1_core_stratixiv_arch_timing.blif.out 2>>
   sparct1_core_stratixiv_arch_timing.blif.err

2. vpr_base.mytest stratixiv_arch.timing.xml
   sparct1_core_stratixiv_arch_timing.blif --place_algorithm bounding_box
   --place_static_notiming_move_prob 50 25 25 --max_criticality 0.0
   --router_initial_timing all_critical --routing_failure_predictor off
   --route_chan_width 300 --max_router_iterations 20 --router_lookahead classic
   --initial_pres_fac 1.0 --pres_fac_mult 2.0 --astar_fac 1.5
   --router_profiler_astar_fac 1.5 --seed 3 --sdc_file
   sparct1_core_stratixiv_arch_timing.sdc --pack_verbosity 0 --netlist_verbosity 0
   --base_cost_type demand_only --place_file
   ref_sparct1_core_stratixiv_arch_timing.place --analysis --route >
   sparct1_core_stratixiv_arch_timing.blif.out 2>>
   sparct1_core_stratixiv_arch_timing.blif.err

```

9. 835.gem5_s

9.1. Test

```

1. gem5sim_base.mytest
   --stats-file=synthetic_traffic.py_LinearGenerator_100.stats.txt
   synthetic_traffic.py LinearGenerator 100 >
   synthetic_traffic.py_LinearGenerator_100.out 2>>
   synthetic_traffic.py_LinearGenerator_100.err

```

9.2. Train

```

1. gem5sim_base.mytest
   --stats-file=synthetic_traffic.py_LinearGenerator_100.stats.txt
   synthetic_traffic.py LinearGenerator 100 >
   synthetic_traffic.py_LinearGenerator_100.out 2>>
   synthetic_traffic.py_LinearGenerator_100.err

```

9.3. Ref

1. `gem5sim_base.mytest --stats-file=bwaves-train-simpt-3-match.py.stats.txt
bwaves-train-simpt-3-match.py > bwaves-train-simpt-3-match.py.out 2>>
bwaves-train-simpt-3-match.py.err`
2. `gem5sim_base.mytest --stats-file=bwaves-train-simpt-3-o3cpu.py.stats.txt
bwaves-train-simpt-3-o3cpu.py > bwaves-train-simpt-3-o3cpu.py.out 2>>
bwaves-train-simpt-3-o3cpu.py.err`

10. 838.diamond_s

10.1. Test

1. `Diamond_base.mytest blastp -q
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz -d
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz --fast --outfmt 6 qseqid
sseqid slen mismatch gapopen qstart qend sstart send nident bitscore -o
astral40.tsv -p 4 > astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz.out 2>>
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz.err`

10.2. Train

1. `Diamond_base.mytest blastp -q
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz -d
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz --mid-sensitive --outfmt 6
qseqid sseqid slen mismatch gapopen qstart qend sstart send nident bitscore -o
astral40.tsv -p 4 > astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz.out 2>>
astral-scopedom-seqres-gd-sel-gs-bib-40-2.07.fa.gz.err`

10.3. Ref

1. `Diamond_base.mytest blastp -q uniprot_sprot.fasta.gz -d uniprot_sprot.fasta.gz
--index-chunks 1 --fast --masking tantan --outfmt 6 qseqid sseqid slen mismatch
gapopen qstart qend sstart send -o uniprot.tsv -p 4 >
uniprot_sprot.fasta.gz.out 2>> uniprot_sprot.fasta.gz.err`
2. `Diamond_base.mytest blastp -q cRAP.fasta.gz -d swissprot.fasta
--ultra-sensitive --index-chunks 4 --block-size 2.8 --masking seg --evaluate
0.0005 --outfmt 6 qseqid sseqid slen mismatch gapopen qstart qend sstart send
-o cRAP.tsv -p 4 > cRAP.fasta.gz.out 2>> cRAP.fasta.gz.err`

11. 846.minizinc_s

11.1. Test

1. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver share/minizinc/solvers/chuffed.msc -s -f radiation.mzn > radiation.mzn.out 2>> radiation.mzn.err`
2. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver gecode_presolver -s tennis_problem.mzn > tennis_problem.mzn.out 2>> tennis_problem.mzn.err`

11.2. Train

1. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver share/minizinc/solvers/chuffed.msc -s -f sudoku_opt.mzn sudoku_p29.dzn > sudoku_p29.dzn.out 2>> sudoku_p29.dzn.err`
2. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver gecode_presolver -s farmer_and_cows.mzn > farmer_and_cows.mzn.out 2>> farmer_and_cows.mzn.err`
3. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver gecode_presolver --solver-statistics roster-sickness.mzn large-2-2.dzn -p 4 > large-2-2.dzn.out 2>> large-2-2.dzn.err`

11.3. Ref

1. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver gecode_presolver --two-pass -s gunport_problem2.mzn > gunport_problem2.mzn.out 2>> gunport_problem2.mzn.err`
2. `minizinc_s_base.mytest --stdlib-dir share/minizinc -r 214365 --output-mode json --solver share/minizinc/solvers/chuffed.msc -f -s yumi-dynamic.mzn > yumi-dynamic.mzn.out 2>> yumi-dynamic.mzn.err`

12. 853.ns3_s

12.1. Test

1. `ns3_s_base.mytest tcp-pacing --isPacingEnabled=true --tracing=false --RngSeed=1 --RngRun=1 > tcp-pacing_0.out 2>> tcp-pacing_0.err`
2. `ns3_s_base.mytest wifi-mixed-network --simulationTime=2 --RngSeed=1 --RngRun=1 > wifi-mixed-network_1.out 2>> wifi-mixed-network_1.err`

12.2. Train

1. `ns3_s_base.mytest wifi-mixed-network --isUdp=0 --simulationTime=10 --RngSeed=1 --RngRun=1 > wifi-mixed-network_0.out 2>> wifi-mixed-network_0.err`
2. `ns3_s_base.mytest wifi-mixed-network --isUdp=1 --simulationTime=10 --payloadSize=2048 --RngSeed=1 --RngRun=1 > wifi-mixed-network_1.out 2>> wifi-mixed-network_1.err`

12.3. Ref

1. `ns3_s_base.mytest traffic-fqcode1 --stopTime=120 --n0TcpType=bic --n1TcpType=reno --bottleneckQueueType=codel --baseRtt=90ms --RngSeed=1 --RngRun=1 > traffic-fqcode1_0.out 2>> traffic-fqcode1_0.err`
2. `ns3_s_base.mytest tcp-pacing --simulationEndTime=750 --useEcn=true --RngSeed=1 --RngRun=1 > tcp-pacing_1.out 2>> tcp-pacing_1.err`

`../run_base_refspped_mytest.0000/ns3_s_base.mytest tcp-validation --stopTime=140s --firstTcpType=cubic --secondTcpType=dctcp --RngSeed=1 --RngRun=1 > tcp-validation_2.out 2>> tcp-validation_2.err`

13. 854.graph500.s

13.1. Test

1. `graph500_s_base.mytest -s 18 -e 16 -o bfs_test.bin -f 1 > bfs_test.bin.out 2>> bfs_test.bin.err`
2. `graph500_s_base.mytest -m 62 -p 100 -q apsp_NYroad_62.bin -g 1 > apsp_NYroad_62.bin.out 2>> apsp_NYroad_62.bin.err`

13.2. Train

1. `graph500_s_base.mytest -s 20 -e 16 -o bfs_train.bin -f 1 > bfs_train.bin.out 2>> bfs_train.bin.err`
2. `graph500_s_base.mytest -m 3508 -p 3000 -q apsp_NYroad_3508.bin -g 1 > apsp_NYroad_3508.bin.out 2>> apsp_NYroad_3508.bin.err`

13.3. Ref

1. `graph500_s_base.mytest -s 26 -e 16 -o bfs_refspeed.bin -f 1 > bfs_refspeed.bin.out 2>> bfs_refspeed.bin.err`
2. `graph500_s_base.mytest -m 19384 -p 16000 -q apsp_NYroad_19384.bin -g 1 > apsp_NYroad_19384.bin.out 2>> apsp_NYroad_19384.bin.err`

SPEC CPU Rate 2026 Floating Point Benchmarks

1. 709.cactus_r

1.1. Test

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

1.2. Train

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

1.3. Ref

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

2. 722.palm_r

2.1. Test

```
1. palm_r_base.mytest < runfile_atmos > palm_r.out 2>> palm_r.err
```

2.2. Train

```
1. palm_r_base.mytest < runfile_atmos > palm_r.out 2>> palm_r.err
```

2.3. Ref

```
1. palm_r_base.mytest < runfile_atmos > palm_r.out 2>> palm_r.err
```

3. 731.astcenc_r

3.1. Test

```
1. astcenc_r_base.mytest test-inputs.txt > astcenc_r.0.out 2>> astcenc_r.0.err
```

3.2. Train

```
1. astcenc_r_base.mytest train-inputs.txt > astcenc_r.0.out 2>> astcenc_r.0.err
```

3.3. Ref

```
1. astcenc_r_base.mytest ref-inputs-linear.txt > astcenc_r.0.out 2>>
   astcenc_r.0.err
2. astcenc_r_base.mytest ref-inputs-hdr.txt > astcenc_r.1.out 2>> astcenc_r.1.err
3. astcenc_r_base.mytest ref-inputs-precision.txt > astcenc_r.2.out 2>>
   astcenc_r.2.err
```

4. 736.ocio_r

4.1. Test

```
1. ocioperf_base.mytest --spec-validation-offset 2 --spec-validation-stride 50
   --spec-validation-pixels 144 --iter 20 --test -1 --transform ctf/metadata.ctf >
   perf_metadata.ctf.out 2>> perf_metadata.ctf.err
```

4.2. Train

```
1. ocioperf_base.mytest --spec-validation-offset 20 --spec-validation-stride 3
   --spec-validation-pixels 128 --bitdepths ui16 f32 --iter 99 --test 0
   --transform ctf/metadata.ctf > perf_metadata.ctf.out 2>> perf_metadata.ctf.err
2. ocioperf_base.mytest --spec-validation-offset 30 --spec-validation-stride 5
   --spec-validation-pixels 129 --bitdepths f32 f32 --iter 30 --test 0 --transform
   clf/aces_to_video_with_look.clf > perf_aces_to_video_with_look.clf.out 2>>
   perf_aces_to_video_with_look.clf.err
3. ocioperf_base.mytest --spec-validation-offset 40 --spec-validation-stride 7
   --spec-validation-pixels 130 --bitdepths ui16 ui16 --iter 22 --8kres --test -1
   --transform clf/lut1d_long.clf > perf_lut1d_long.clf.out 2>>
   perf_lut1d_long.clf.err
4. ocioperf_base.mytest --spec-validation-offset 50 --spec-validation-stride 9
   --spec-validation-pixels 131 --bitdepths ui16 ui16 --iter 2 --8kres --test -1
   --transform clf/heavy_transform.clf > perf_heavy_transform.clf.out 2>>
   perf_heavy_transform.clf.err
```

4.3. Ref

1. `ocioperf_base.mytest --spec-validation-offset 101 --spec-validation-stride 17 --spec-validation-pixels 131 --bitdepths ui16 ui16 --iter 100 --test -1 --transform ctf/lut1d_halfdom.ctf > perf_lut1d_halfdom.ctf.out 2>> perf_lut1d_halfdom.ctf.err`
2. `ocioperf_base.mytest --spec-validation-offset 202 --spec-validation-stride 19 --spec-validation-pixels 132 --bitdepths ui16 f32 --iter 200 --8kres --test 0 --transform ctf/mntr_srgb_identity.ctf > perf_mntr_srgb_identity.ctf.out 2>> perf_mntr_srgb_identity.ctf.err`

5. 737.gmsh_r

5.1. Test

1. `gmsh_r_base.mytest -option gmsh.opts -nt 0 spec.geo -smooth 0 -clscale .60 > spec.geo.out 2>> spec.geo.err`

5.2. Train

1. `gmsh_r_base.mytest -option gmsh.opts -nt 0 sphere-discrete.geo > sphere-discrete.geo.out 2>> sphere-discrete.geo.err`
2. `gmsh_r_base.mytest -option gmsh.opts -nt 0 spec.geo -smooth 2 -clscale .27 > spec.geo.out 2>> spec.geo.err`

5.3. Ref

1. `gmsh_r_base.mytest -option gmsh.opts -nt 0 choi.geo > choi.geo.out 2>> choi.geo.err`
2. `gmsh_r_base.mytest -option gmsh.opts -nt 0 mediterranean.geo > mediterranean.geo.out 2>> mediterranean.geo.err`

6. 748.flightdm_r

6.1. Test

1. `JSBSim_base.mytest --nohighlight scripts/c1722.xml > c1722.xml.out 2>> c1722.xml.err`
2. `JSBSim_base.mytest --nohighlight scripts/J2461.xml > J2461.xml.out 2>> J2461.xml.err`

6.2. Train

```
1. JSBSim_base.mytest --nohighlight scripts/weather-balloon.xml >
   weather-balloon.xml.out 2>> weather-balloon.xml.err
2. JSBSim_base.mytest --nohighlight scripts/f16_test.xml > f16_test.xml.out 2>>
   f16_test.xml.err
3. JSBSim_base.mytest --nohighlight scripts/c3105.xml > c3105.xml.out 2>>
   c3105.xml.err
4. JSBSim_base.mytest --nohighlight scripts/Short_S23_2.xml > Short_S23_2.xml.out
   2>> Short_S23_2.xml.err
```

6.3. Ref

```
1. JSBSim_base.mytest --nohighlight scripts/weather-balloon2.xml >
   weather-balloon2.xml.out 2>> weather-balloon2.xml.err
2. JSBSim_base.mytest --nohighlight scripts/B747_script1.xml >
   B747_script1.xml.out 2>> B747_script1.xml.err
```

7. 749.fotonik3d_r

7.1. Test

```
1. fotonik3d_r_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

7.2. Train

```
1. fotonik3d_r_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

7.3. Ref

```
1. fotonik3d_r_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

8. 765.roms_r

8.1. Test

```
1. roms_r_base.mytest < roms_benchmark0.in.x > roms_benchmark0.log 2>>
   roms_benchmark0.err
```

8.2. Train

```
1. roms_r_base.mytest < roms_benchmark1.in.x > roms_benchmark1.log 2>>
   roms_benchmark1.err
```

8.3. Ref

```
1. roms_r_base.mytest < roms_benchmark2.in.x > roms_benchmark2.log 2>>
   roms_benchmark2.err
```

9. 766.femflow_r

9.1. Test

```
1. femflow_r_base.mytest test.prm > femflow.out 2>> femflow.err
```

9.2. Train

```
1. femflow_r_base.mytest train.prm > femflow.out 2>> femflow.err
```

9.3. Ref

```
1. femflow_r_base.mytest refrate.prm > femflow.out 2>> femflow.err
```

10. 767.nest_r

10.1. Test

```
1. nest_r_base.mytest BrodyHopfield.sli > BrodyHopfield.sli.out 2>>
   BrodyHopfield.sli.err
2. nest_r_base.mytest balancedneuron.sli > balancedneuron.sli.out 2>>
   balancedneuron.sli.err
```

10.2. Train

```
1. nest_r_base.mytest balancedneuron-2.sli > balancedneuron-2.sli.out 2>>
   balancedneuron-2.sli.err
2. nest_r_base.mytest cuba_ps.sli > cuba_ps.sli.out 2>> cuba_ps.sli.err
3. nest_r_base.mytest cuba.sli > cuba.sli.out 2>> cuba.sli.err
```

10.3. Ref

```
1. nest_r_base.mytest cuba_stdp.sli > cuba_stdp.sli.out 2>> cuba_stdp.sli.err
2. nest_r_base.mytest structural_plasticity_benchmark.sli >
   structural_plasticity_benchmark.sli.out 2>>
   structural_plasticity_benchmark.sli.err
```

11. 772.marian_r

11.1. Test

```
1. marian-decoder_base.mytest --cpu-threads 1 -m model.alphas.npz -v vocab.spm
   vocab.spm --beam-size 1 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
   -w 64 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
   standard-only --quiet --quiet-translation -i TildeMODEL-test.en --log
   TildeMODEL-test.log --log-level off -o TildeMODEL-test.out >
   run_TildeMODEL-test.out.out 2>> run_TildeMODEL-test.out.err
```

11.2. Train

```
1. marian-decoder_base.mytest --cpu-threads 1 -m model.alphas.npz -v vocab.spm
   vocab.spm --beam-size 2 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
   -w 64 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
   standard-only --quiet --quiet-translation -i technology_and_courage.en --log
   technology_and_courage.log --log-level off -o technology_and_courage.out >
   run_technology_and_courage.out.out 2>> run_technology_and_courage.out.err
```

11.3. Ref

```
1. marian-decoder_base.mytest --cpu-threads 1 -m model.alphas.npz -v vocab.spm
   vocab.spm --beam-size 1 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
   -w 512 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
   standard-only --quiet --quiet-translation -i TildeMODEL-spec.en --log
   TildeMODEL-spec.log --log-level off -o TildeMODEL-spec.out >
   run_TildeMODEL-spec.out.out 2>> run_TildeMODEL-spec.out.err

2. marian-decoder_base.mytest --cpu-threads 1 -m model.alphas.npz -v vocab.spm
   vocab.spm --beam-size 1 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
   -w 512 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
   standard-only --quiet --quiet-translation -i EuroPat-spec.en --log
   EuroPat-spec.log --log-level off -o EuroPat-spec.out > run_EuroPat-spec.out.out
   2>> run_EuroPat-spec.out.err
```

12. 782.lbm_r

12.1. Test

```
1. lbm_r_base.mytest 10 reference.dat 0 0 200_200_130_ldc.of > lbm.out 2>> lbm.err
```

12.2. Train

```
1. lbm_r_base.mytest 120 reference.dat 0 0 200_200_130_ldc.of > lbm.out 2>>  
lbm.err
```

12.3. Ref

```
1. lbm_r_base.mytest 900 reference.dat 0 0 200_200_130_ldc.of > lbm.out 2>>  
lbm.err
```


SPEC CPU Speed 2026 Floating Point Benchmarks

1. 800.pot3d_s

1.1. Test

```
1. pot3d_base.mytest 1 > pot3d.stdout.out 2>> pot3d.stderr.err
```

1.2. Train

```
1. pot3d_base.mytest 1 > pot3d.stdout.out 2>> pot3d.stderr.err
```

1.3. Ref

```
1. pot3d_base.mytest 1 > pot3d.stdout.out 2>> pot3d.stderr.err
```

2. 803.sph_exa_s

2.1. Test

```
1. sph_exa_base.mytest -n 100 -s 0 -w 1 > sph_exa.out 2>> sph_exa.err
```

2.2. Train

```
1. sph_exa_base.mytest -n 50 -s 7 -w 5 > sph_exa.out 2>> sph_exa.err
```

2.3. Ref

```
1. sph_exa_base.mytest -n 210 -s 24 -w 7 > sph_exa.out 2>> sph_exa.err
```

3. 809.cactus_s

3.1. Test

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

3.2. Train

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

3.3. Ref

```
1. cactus_base.mytest ShiftedGaugeWave.par > cactus.out 2>> cactus.err
```

4. 811.tealeaf_s

4.1. Test

```
1. tealeaf_base.mytest > tea.stdout 2>> tea.err
```

4.2. Train

```
1. tealeaf_base.mytest > tea.stdout 2>> tea.err
```

4.3. Ref

```
1. tealeaf_base.mytest > tea.stdout 2>> tea.err
```

5. 816.nab_s

5.1. Test

```
1. nab_s_base.mytest --directory hkrdenq --seed 1930344093 --iterations 1000 >
   hkrdenq.1000.out 2>> hkrdenq.1000.err
```

5.2. Train

```
1. nab_s_base.mytest --directory aminos --seed 391519156 --iterations 1000 >
   aminos.1000.out 2>> aminos.1000.err

2. nab_s_base.mytest --directory gcn4dna --seed 1850041461 --iterations 300 >
   gcn4dna.300.out 2>> gcn4dna.300.err
```

5.3. Ref

```
1. nab_s_base.mytest --directory 3j1n --seed 20140317 --iterations 220 --epoques 3
   > 3j1n.220.3.out 2>> 3j1n.220.3.err

2. nab_s_base.mytest --directory 1am0 --seed 20220406 --iterations 500 --epoques 3
   > 1am0.500.3.out 2>> 1am0.500.3.err
```

6. 820.cloverleaf_s

6.1. Test

```
1. cloverleaf_s_base.mytest > clover_leaf.log 2>> clover_leaf.err
```

6.2. Train

```
1. cloverleaf_s_base.mytest > clover_leaf.log 2>> clover_leaf.err
```

6.3. Ref

```
1. cloverleaf_s_base.mytest > clover_leaf.log 2>> clover_leaf.err
```

7. 822.palm_s

7.1. Test

```
1. palm_base.mytest < runfile_atmos > palm.out 2>> palm.err
```

7.2. Train

```
1. palm_base.mytest < runfile_atmos > palm.out 2>> palm.err
```

7.3. Ref

```
1. palm_base.mytest < runfile_atmos > palm.out 2>> palm.err
```

8. 849.fotonik3d_s

8.1. Test

```
1. fotonik3d_s_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

8.2. Train

```
1. fotonik3d_s_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

8.3. Ref

```
1. fotonik3d_s_base.mytest > fotonik3d.log 2>> fotonik3d.err
```

9. 857.namd_s

9.1. Test

```
1. namd_s_base.mytest --input stmv.input --iterations 1 --output stmv.test.output  
  > namd.out 2>> namd.err
```

9.2. Train

```
1. namd_s_base.mytest --input stmv.input --iterations 49 --output  
  stmv.train.output > namd.out 2>> namd.err
```

9.3. Ref

```
1. namd_s_base.mytest --input stmv.input --output stmv.ref.output --iterations 399  
  > namd.out 2>> namd.err
```

10. 865.roms_s

10.1. Test

```
1. roms_s_base.mytest < roms_benchmark0.in > roms_benchmark0.log 2>>  
  roms_benchmark0.err
```

10.2. Train

```
1. roms_s_base.mytest < roms_benchmark1.in > roms_benchmark1.log 2>>  
  roms_benchmark1.err
```

10.3. Ref

```
1. roms_s_base.mytest < roms_benchmark3.in > roms_benchmark3.log 2>>  
  roms_benchmark3.err
```

11. 867.nest_s

11.1. Test

```
1. nest_s_base.mytest --userargs=threads=4 --verbosity=FATAL cuba_stdp_spec.sli >
  cuba_stdp_spec.sli.out 2>> cuba_stdp_spec.sli.err
```

11.2. Train

```
1. nest_s_base.mytest --userargs=threads=4 --verbosity=FATAL cuba_stdp_spec.sli >
  cuba_stdp_spec.sli.out 2>> cuba_stdp_spec.sli.err

2. nest_s_base.mytest --userargs=threads=4 --verbosity=FATAL
  brunel-2000_newconnect_dc.sli > brunel-2000_newconnect_dc.sli.out 2>>
  brunel-2000_newconnect_dc.sli.err
```

11.3. Ref

```
1. nest_s_base.mytest --userargs=threads=4 --verbosity=FATAL microcircuit_spec.sli
  > microcircuit_spec.sli.out 2>> microcircuit_spec.sli.err

2. nest_s_base.mytest --userargs=threads=4 --verbosity=FATAL
  hpc_benchmark_spec.sli > hpc_benchmark_spec.sli.out 2>>
  hpc_benchmark_spec.sli.err
```

12. 872.marian_s

12.1. Test

```
1. marian-decoder_base.mytest --cpu-threads 4 -m model.alphas.npz -v vocab.spm
  vocab.spm --beam-size 1 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
  -w 64 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
  standard-only --quiet --quiet-translation -i TildeMODEL-test.en --log
  TildeMODEL-test.log --log-level off -o TildeMODEL-test.out >
  run_TildeMODEL-test.out.out 2>> run_TildeMODEL-test.out.err
```

12.2. Train

```
1. marian-decoder_base.mytest --cpu-threads 4 -m model.alphas.npz -v vocab.spm
  vocab.spm --beam-size 2 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
  -w 64 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
  standard-only --quiet --quiet-translation -i technology_and_courage.en --log
  technology_and_courage.log --log-level off -o technology_and_courage.out >
  run_technology_and_courage.out.out 2>> run_technology_and_courage.out.err
```

12.3. Ref

```
1. marian-decoder_base.mytest --cpu-threads 4 -m model.alphas.npz -v vocab.spm
vocab.spm --beam-size 8 --mini-batch 32 --maxi-batch 100 --maxi-batch-sort src
-w 64 --skip-cost --gemm-type intgemm8 --intgemm-options precomputed-alpha
standard-only --quiet --quiet-translation -i TildeMODEL-spec-long.en --log
TildeMODEL-spec.log --log-level off -o TildeMODEL-spec.out >
run_TildeMODEL-spec.out.out 2>> run_TildeMODEL-spec.out.err
```

13. 881.neutron.s

13.1. Test

```
1. neutron_base.mytest xsbench -s small -g 1000 -p 50000 > xsbench.small.out 2>>
xsbench.small.err

2. neutron_base.mytest rsbench -m event -l 100000 -p 2000 -P 2000 -d >
rsbench.event.out 2>> rsbench.event.err
```

13.2. Train

```
1. neutron_base.mytest xsbench -G hash -h 100 -p 100000 -g 500000 -s small >
xsbench.hash.out 2>> xsbench.hash.err

2. neutron_base.mytest rsbench -m history -l 1000 -p 1000 -P 1000 -W 100 -d >
rsbench.history.out 2>> rsbench.history.err
```

13.3. Ref

```
1. neutron_base.mytest xsbench -G unionized -m event -s large -l 800000000 -g
82500 -k 1 > xsbench.unionized.out 2>> xsbench.unionized.err

2. neutron_base.mytest rsbench -m history -s large -p 1500000 -P 300000 -W 35000
-l 60 -d > rsbench.history.out 2>> rsbench.history.err
```

specrand_r and specrand_s Benchmarks

1. 999.specrand_r

1.1. Test

```
1. specrand_r_base.mytest 324342 24239 > rand.specrand.24239.out 2>>
   rand.specrand.24239.err

2. distributions_r_base.mytest 324342 24239 > rand.distributions.24239.out 2>>
   rand.distributions.24239.err
```

1.2. Train

```
1. specrand_r_base.mytest 1 11 > rand.specrand.11.out 2>> rand.specrand.11.err

2. distributions_r_base.mytest 1 11 > rand.distributions.11.out 2>>
   rand.distributions.11.err
```

1.3. Ref

```
1. specrand_r_base.mytest 1255432124 234923 > rand.specrand.234923.out 2>>
   rand.specrand.234923.err

2. distributions_r_base.mytest 1255432124 234923 > rand.distributions.234923.out
   2>> rand.distributions.234923.err
```

2. 998.specrand_s

2.1. Test

```
1. specrand_s_base.mytest 324342 24239 > rand.specrand.24239.out 2>>
   rand.specrand.24239.err

2. distributions_s_base.mytest 324342 24239 > rand.distributions.24239.out 2>>
   rand.distributions.24239.err
```

2.2. Train

```
1. specrand_s_base.mytest 1 11 > rand.specrand.11.out 2>> rand.specrand.11.err

2. distributions_s_base.mytest 1 11 > rand.distributions.11.out 2>>
   rand.distributions.11.err
```

2.3. *Ref*

1. `specrand_s_base.mytest 1255432124 234923 > rand.specrand.234923.out 2>>
rand.specrand.234923.err`
2. `distributions_s_base.mytest 1255432124 234923 > rand.distributions.234923.out
2>> rand.distributions.234923.err`

References

- [1] <https://www.spec.org/cpu2026/Docs/>